Weekly Report 2

1. Time spent: 6 hours
2. What did I do this week?
    a. Researched the Markov chain topic further, with a focus on understanding how to implement sliding windows
    b. Experimented with different data structures before settling on a combination of dictionaries and tuples to represent states and transitions.
    c. Researched unit testing and thought about which parts of each method needs to be tested.
3. How has the program progressed?
    a. The core code is functional. It is able to read, clean and tokenise the articles, and a text of a given length can be produced.
    b. The function ‘clean_ articles’ has been unit tested to determine whether it is able to handle uppercase letters, html tags and urls.
4. What did I learn this week/today?
    a. How sliding windows are used to extract n-gram states from a list of tokens.
    b. What makes a good unit test, and how to identify the most important parts to test in a function.
5. What remains unclear or has been challenging?
    a. The generated text is, in many cases, awkwardly worded and doesn’t make sense.
    b. I am not sure if the model will be able to produce text that resembles a news article, given that its text generation is based solely on probability rather than grammar, sentence structure etc.
6. What will I do next?
    a. Improve and expand on the unit testing, particularly for the model and text generation functions.
    b. Make improvements on text preprocessing functions which will hopefully make for a better input for the model.
    c. Begin working on a user interface.